iLearn Practice Exam

10BR · Chapter / Topic 5

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Chapter 5: The Internet and its Uses

Name: _____

Class: _____

Date: _____

Time allowed: 1 hour 30 minutes

TOTAL MARKS: **75**

Marks obtained: ..... / 75

INSTRUCTIONS

· Answer all questions.
· Write your answers in the spaces provided on this paper, using a black or dark blue pen.
· Do not use an erasable pen or correction fluid.
· No calculators are allowed.
· Do not write anything in the margins.

INFORMATION

· The total mark for this paper is 75.
· The number of marks for each question or part question is shown in brackets [ ].
· This paper contains 9 questions, all drawn from Cambridge past papers on this chapter.
· After you finish, your teacher will mark your work against the official mark scheme.

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1 A student is writing a help guide about how to recognise and avoid the cyber-security threat of pharming.

(a) Give **three** appropriate solutions he could include.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

(b) The student also wants to include information in the help guide about the use of social engineering as a cyber-security threat.

Describe what is meant by social engineering.

Include **one** example of social engineering in your answer.

.....

.....

.....

.....

.....

..... [3]

(c) The student includes information about the security solution of access levels.

Describe what is meant by access levels.

.....

.....

.....

.....

.....

..... [3]

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2 The table contains terms and descriptions about the internet.

Complete the table with the missing terms and descriptions.

Term	Description
.....	the collective name for all the web pages available
.....	a small text file, stored by the web browser, that can store a user's personal data
uniform resource locator (URL)	....
web server	....
.....	the language used to create a website. Example tags are <head> and <body>
.....	a protocol that is used to request and send web pages

[6]

3 A student enters a uniform resource locator (URL) into their web browser.

Draw and annotate a diagram to show the process from the user entering the URL to the web browser displaying the web page.

[5]

4 A company has a website that is suffering a distributed denial of service (DDoS) attack.

(a) Draw and annotate a diagram to show the process of the DDoS.

[5]

(b) Identify a solution that can be used to help prevent the DDoS attack being successful.

..... [1]

5 Complete the table to give the missing term or description for the internet terms.

Internet term	Description
.....	the collection of web pages accessed using the internet
.....	the address given to a device when it connects to the internet
web browser	....
.....	the hardware that stores a database of matching website and IP addresses
.....	a type of hardware that can be used to prevent a distributed denial of service (DDoS) attack
hacking	....

[6]

6 An online company has a website that allows a user to buy products.

The user types in the uniform resource locator (URL) for the website and this is sent to a domain name server (DNS).

(a) Describe the role of the DNS.

.....

.....

.....

..... [2]

(b) The website asks a user to input their username and password to log into their account.

The company want to make their login system more secure.

Tick (✓) **one** box to show **one** other security method that could be added to make the login system more secure.

A access levels ☐

B firewall ☐

C spyware ☐

D two-step verification ☐

[1]

(c) The company uses a proxy server to keep the web server safe from distributed denial of service (DDoS) attacks.

(i) Complete and annotate a diagram to show how a DDoS attack is carried out on the web server.



[5]

(ii) Identify **two** tasks that can be performed by a proxy server that are **not** performed by a firewall.

1 .....

.....

2 .....

.....

[2]

(d) The company uses cookies to allow users to keep items in a shopping cart. The items remain in the cart after the user closes their web browser.

(i) Tick (✓) **one** box to show the type of cookie that is used for this process.

A dynamic ☐

B persistent ☐

C session ☐

D static ☐

[1]

(ii) Give **three** other ways that the company can use cookies.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

(iii) Cookies are stored by a web browser.

Give **two** other functions of a web browser.

1 .....

.....

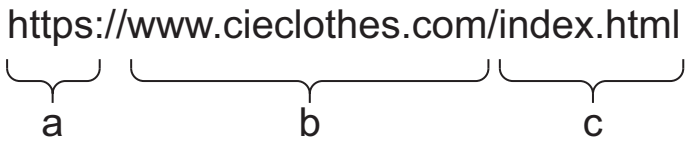
2 .....

.....

[2]

7 An online clothing company has a homepage at the uniform resource locator (URL) https://www.cieclothes.com/index.html

(a) Write down the **three** different parts of the URL shown by the labels a, b and c.



a ..... b ..... c .....

[3]

(b) The web pages are stored on a web server. A proxy server is used to stop distributed denial of service (DDoS) attacks on the web server.

Describe how the proxy server can stop a DDoS attack reaching a web server.

.....

.....

.....

.....

.....

..... [3]

(c) Customers use the website to buy clothing from the company.

Complete the table with the missing terms and descriptions about retrieving and displaying web pages.

Term	Description
hypertext markup language (HTML)	....
domain name server (DNS)	....
.....	It is a numerical address used to locate the web server.
.....	It is a secure protocol that can be used to transmit data sent to and received from a web server.
.....	It is a piece of software that is used to display web pages.

[5]

(d) The clothing company uses cookies to store customer details.

Complete the statements about cookies.

Use terms from the list. **Not** all terms need to be used. You should only use a term once.

asymmetric DNS dynamic persistent session

static symmetric user utility software web browser

Cookies are stored and managed by the ..... .

..... cookies are temporary files that are created when a user visits a web page.

..... cookies are permanent files that are created when a user visits a web page.

[3]

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8 Web pages need to be located and retrieved to be displayed on a device.

(a) Complete the paragraph about locating and retrieving web pages.

Use the terms from the list. Some of the terms in the list will **not** be used.

You should only use a term once.

certificate	database
domain name server (DNS)	encryption
hypertext markup language (HTML)	internet protocol (IP) address
low-level language	protocol
request	uniform resource locator (URL)
web browser	web server

The user types the URL into the address bar. The web browser sends the domain name from the URL to the ..... .

This searches for the matching ..... and sends this back to the ..... .

The web browser sends a ..... to the web server for the web page. The web server then sends the data for the web page back to the web browser. The web browser renders the ..... to display the web page.

[5]

(b) The transmission of data between the web browser on a device and a web server can be secured using the secure socket layer (SSL) protocol.

Complete and annotate the diagram to show the process of sending the data using SSL.



[5]

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9 A student develops a program for their computer science project using a high-level language.

(a) The student uses a web browser to upload their program to a web page. They bookmark the web page where they have uploaded their program, to make it easier to find the web page again.

(i) Give **one** other function of a web browser that the student could use to find the web page again.

..... [1]

(ii) The web page has a uniform resource locator (URL).

Identify **two** parts of a URL.

1 .....

2 .....
[2]

END OF QUESTION PAPER